

Homework #2: Process Simulation and ACF/PACF Analysis

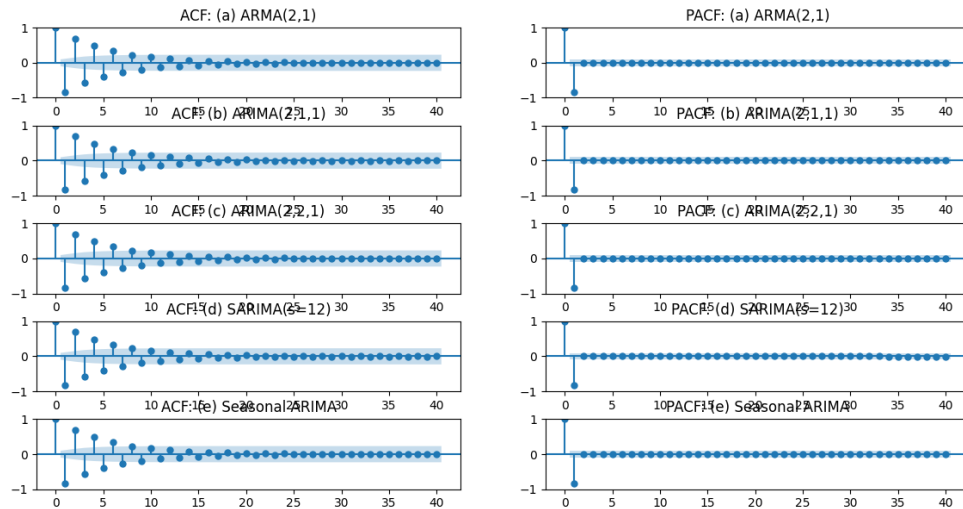
1. Definition of Time Series Processes

As specified in the assignment, we define the following processes using the backshift operator B , where $w_t \sim WN(0, 1)$.

- (a) ARMA(2, 1): $(1 + 0.7B - 0.6B^2)x_t = (1 - 0.5B)w_t$
- (b) ARIMA(2, 1, 1): $(1 + 0.7B - 0.6B^2)(1 - B)x_t = (1 - 0.5B)w_t$
- (c) ARIMA(2, 2, 1): $(1 + 0.7B - 0.6B^2)(1 - B)^2x_t = (1 - 0.5B)w_t$
- (d) SARIMA(2, 0, 1) $\times$ (0, 1, 0)₁₂: $(1 + 0.7B - 0.6B^2)(1 - B^{12})x_t = (1 - 0.5B)w_t$
- (e) SARIMA(2, 1, 1) $\times$ (0, 1, 0)₁₂: $(1 + 0.7B - 0.6B^2)(1 - B)(1 - B^{12})x_t = (1 - 0.5B)w_t$

2. ACF and PACF Visualization

The following Python script simulates these processes and generates the required plots. For non-stationary processes (b, c, d, e), the ACF is expected to decay very slowly.



```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
4
5 def simulate_hw2(n=400):
6     np.random.seed(42)
7     w = np.random.normal(0, 1, n + 200)
8     x_a = np.zeros(n + 200)
9     # Generate ARMA(2,1) core
10    for t in range(2, n + 200):
11        x_a[t] = -0.7*x_a[t-1] + 0.6*x_a[t-2] + w[t] - 0.5*w[t-1]
12
13    # Apply Integrations and Seasonality
14    x_a_cut = x_a[200:]
15    x_b = np.cumsum(x_a_cut) # ARIMA(2,1,1)
16    x_c = np.cumsum(x_b)    # ARIMA(2,2,1)
17
18    # Seasonal (s=12)
19    x_d = np.zeros(len(x_a_cut))
20    for t in range(12, len(x_a_cut)):
21        x_d[t] = x_d[t-12] + x_a_cut[t]
22
23    x_e = np.cumsum(x_d)
24    return [x_a_cut, x_b, x_c, x_d, x_e]

```

Listing 1: Python Simulation Script

3. Analysis of Model Identification

Separating ARMA, ARIMA, and SARIMA:

- **ARMA:** Identified by a rapidly decaying ACF (Stationarity).
- **ARIMA:** Identified by an ACF that stays high and decays very slowly, indicating a unit root and the need for differencing d .
- **SARIMA:** Identified by significant spikes at seasonal lags (e.g., $h = 12, 24$) in the ACF/PACF.

Estimating Orders (p, q, d, P, Q, D, S) :

- **d and D :** Determined by the number of differencing steps required to achieve a stationary ACF.
- **p and q :** After differencing, the PACF "cuts off" at lag p for AR processes, and the ACF "cuts off" at lag q for MA processes.
- **Seasonal P, Q :** Similar cut-off rules apply at seasonal lags $S, 2S, \dots$.

References

- [1] ARMA Model Basics, 2102-503 Lecture Slides.
- [2] ARMA Estimator (Forecasting), 2102-503 Lecture Slides.
- [3] Yule-Walker Estimation and PACF, 2102-503 Lecture Slides.
- [4] Homework Assignment #2 Description.